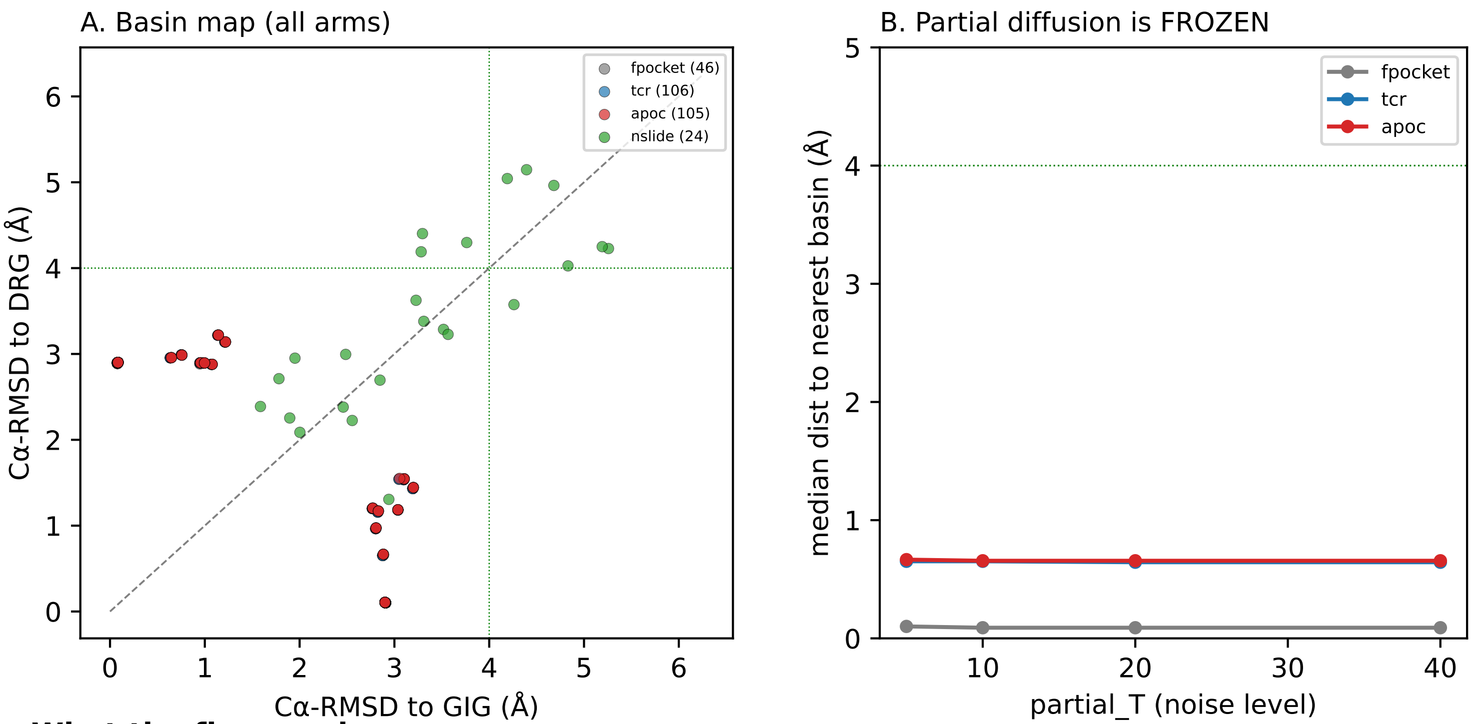
V3 — does interaction conditioning steer the peptide basin?

Peptide-only RFdiffusion in a fixed pMHC-TCR complex; read-out = basin geometry, floor frame.



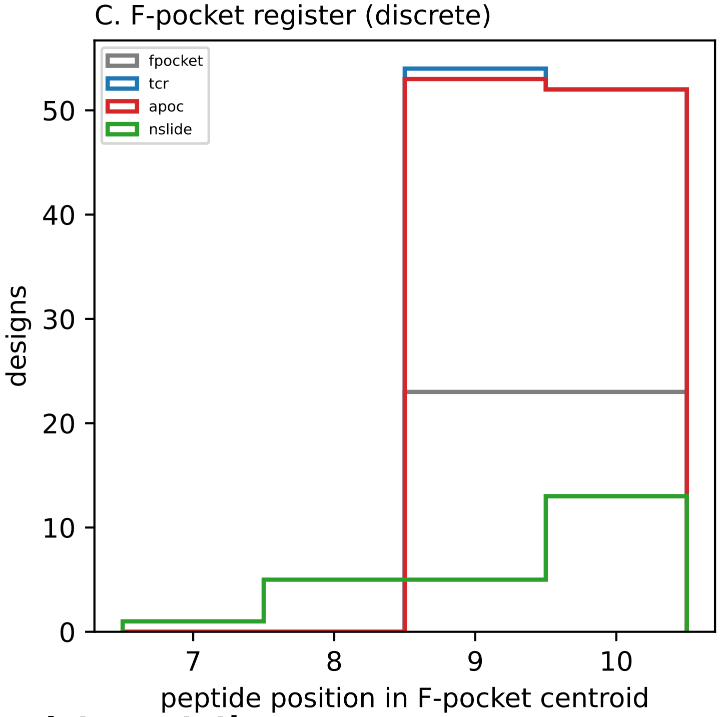
What the figures show

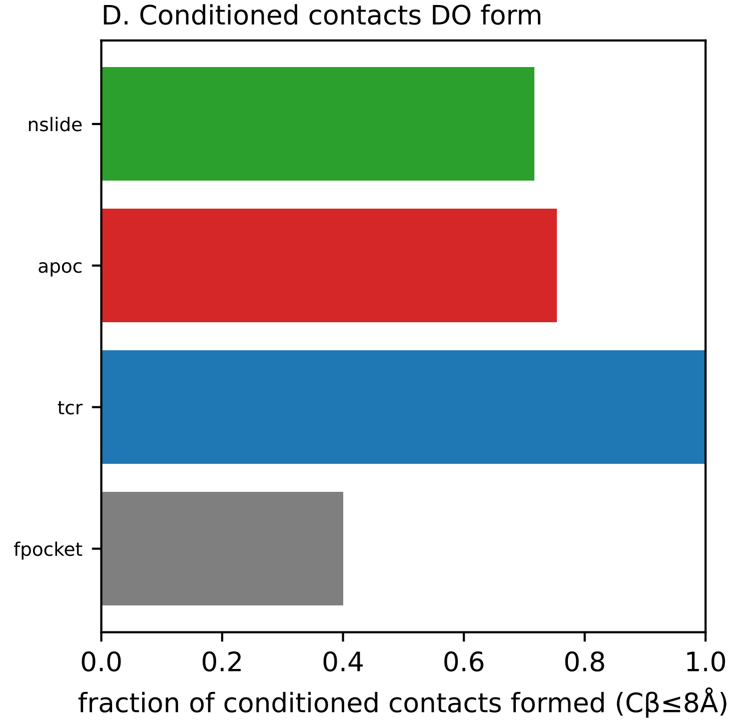
• The two native registers (GIG=6AM5 P10-anchor, DRG=6AMU P9-anchor) are only 2.91 Å apart in the floor frame but geometrically distinct (F-pocket P9↔P10 swap).
• Panel A: every partial-diffusion design (fpocket/tcr/apoc) sits essentially ON a native basin — and always its OWN (cognate) basin. Register transfer across all partial arms = 0 / 257 designs. No steering.
• Panel B: increasing the noise from partial_T=5→40 does NOT move the peptide off native — the curve is flat (median 0.1–0.65 Å to the seed's native basin at every T; 100% 'seated'). The fixed MHC+TCR context + provide_seq over-constrains the peptide, which denoises straight back to its starting register regardless of noise. partial_T ≈50 is full diffusion, so ~40 is already near the ceiling.
• Conditioning on the interactions (Q30α, CDR3β-Phe, A/B-pocket, floor F-pocket) — applied JOINTLY per arm — therefore never gets a chance to act in partial mode: the backbone doesn't leave native.

VERDICT (partial arms): interaction conditioning does not steer the basin here — it is a null result driven by an over-constrained generator, not by the conditioning being wrong.

V3 — mechanism: a discrete register, and no bridge between the two

Why nothing interpolates GIG↔DRG, and confirmation that the conditioned contacts DO form.





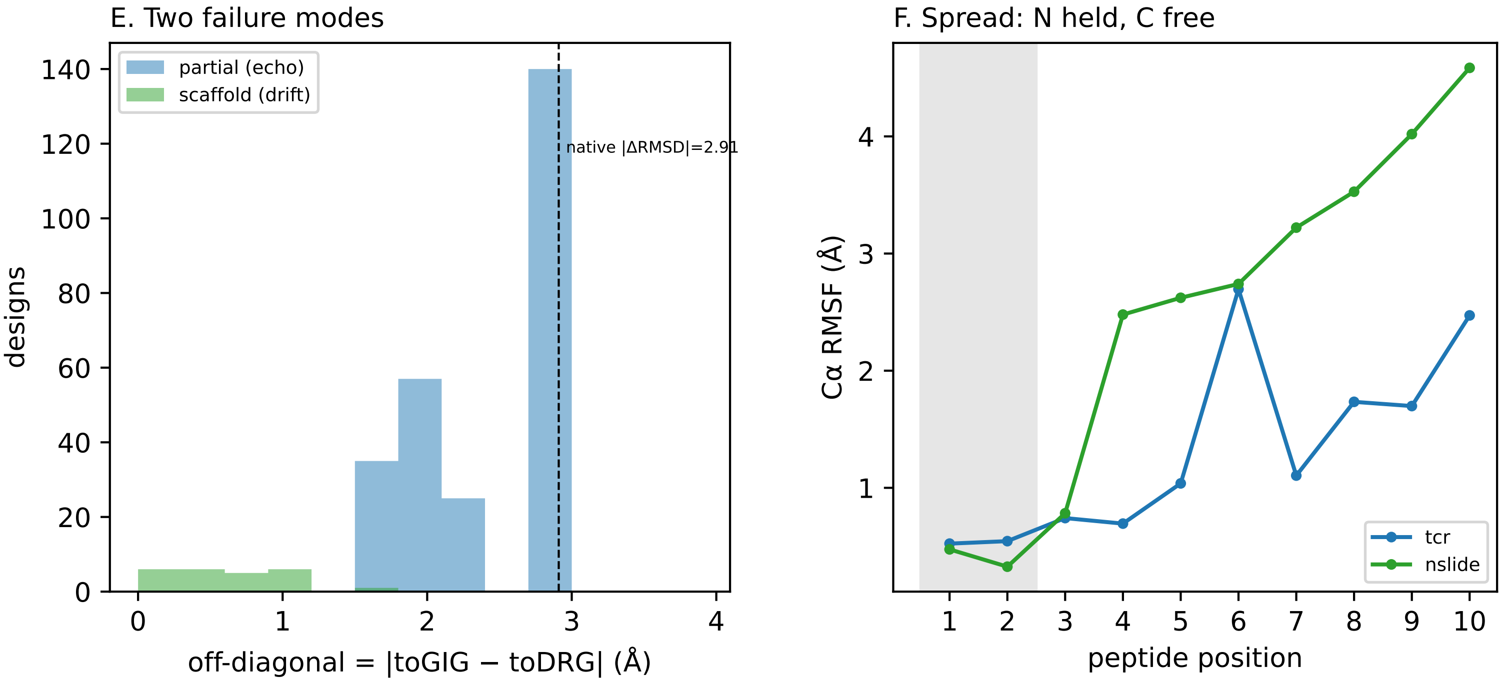
Interpretation

• Panel C — the register is DISCRETE. The peptide Cα occupying the F-pocket centroid is either P10 (GIG-like, unshifted) or P9 (DRG-shifted); no design occupies an intermediate. The GIG↔DRG difference is a two-state P9↔P10 swap, not a continuous deformation — there is no low-RMSD intermediate backbone to design into.
• NO BRIDGE. Zero partial designs cross to the non-cognate basin; the scaffold arms that DO leave native drift to a register-ambiguous middle (~equidistant, ~3 Å from both) — that is under-determination, not a stable connecting state. (A naive 'within 4 Å of both basins' count is meaningless here because the basins are only 2.91 Å apart — the threshold exceeds the separation.)
• Panel D — the conditioning IS being satisfied at the backbone level: the TCR contacts (Q30α + CDR3β-Phe) form in ~100% of tcr designs; the A/B-pocket arm ~0.76; the floor F-pocket reads lower (~0.40) only because those contacts are sidechain-mediated and invisible at the Cβ level on backbone-only designs. So the peptide forms the contacts we asked for — it just does so while staying in its native register.

TAKEAWAY: satisfying the conditioned interactions is necessary but not sufficient to move register; register is selected at the C-terminal F-pocket, which the fixed context pins.

V3 — two failure modes, methods, and verdict

Partial (over-constrained) vs scaffold (under-constrained); what was run; what it means.



Methods (verified against coordinates)

• Seeds (17): 2 RAW crystal (6AM5, 6AMU — no MD) + 15 OpenMM-relaxed snapshots; 6AMT (unshifted DRG) held out as a reference only. Trimmed to α1α2 + peptide + TCR variable (all hotspots retained; verified no interface residue clipped). α2 arm His145–Val152 left free to breathe.
• Conditioning is applied JOINTLY per arm (one ppi.hotspot_res list, one run): fpocket=[D30,A77,A84,A116,A123]; tcr=[D30,E97|E100]; apoc=tcr+[A7,A63,A66,A70,A99,A159,A171]; nslide/sc=scaffold rebuild. E97(6AM5)/E100(6AMU) is the epitope-specific CDR3 β-Phe.
• Frame: superpose on 61 β-sheet-floor Cα (helices, TCR, arm 145–152 excluded); 6AM5↔6AMU floor RMSD = 0.21 Å — the invariant frame that makes the basin map interpretable. Peptide RMSD to 3 refs (GIG / DRG-shift / DRG-unshift).
• Geometry QC: 0 broken designs (Cα-Cα bonds + clashes) among those shown.

Verdict & next

• Panel E: partial arms pile at the native |ΔRMSD| (off-diagonal ≈ native, i.e. IN a basin but only because they never left native); scaffold arms collapse toward 0 (on the diagonal = ambiguous drift). Neither selects a NEW basin.
• Panel F: RMSF is low at the conditioned N-terminus (P1-2) and high at the free C-terminus — the backbone spread lives exactly where the register is decided, i.e. where conditioning has least grip.
• BOTTOM LINE: interaction-only conditioning did not steer or bridge the register. Partial diffusion is over-constrained (echoes native at all T); scaffold rebuild is under-constrained (drifts, no stable intermediate). The two registers behave as separate discrete wells. NEXT: seed from real MD-excursion / mid-transition backbones (cached dual-RMSD trajectory) rather than near-native crystal/relaxed frames — give diffusion a non-native starting point to bridge from.